

Immediate Auditory Feedback Speeds Visuomotor Delay Adaptation in Video Games

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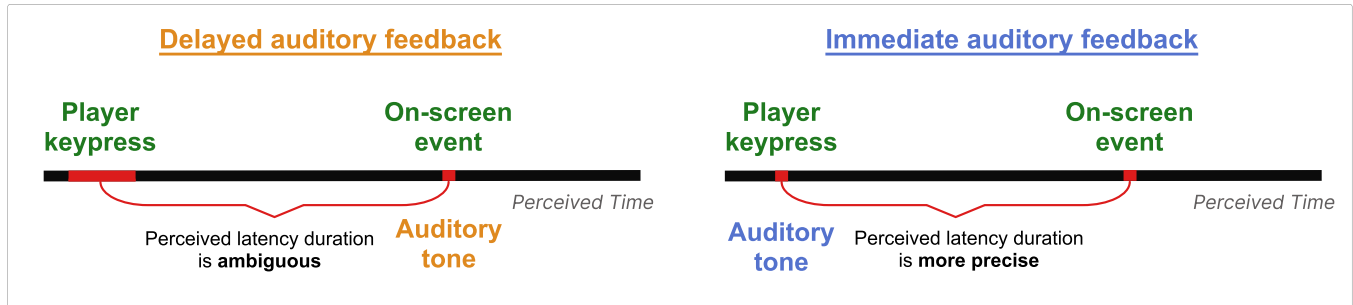


Figure 1: Illustration of how immediate (relative to delayed) auditory feedback increases the precision of perceived keypress timing, and thus the precision with which visuomotor latency is perceived.

Abstract

When latency occurs in online games, players must adapt by entering inputs earlier. But when pressing a key, it is hard to know exactly when it was registered. Here we hypothesized that immediate auditory feedback helps players to perceive precisely when input was registered — allowing for more precise measurement of latency, and faster visuomotor adaptation. To test this, we designed two browser-based games requiring time-sensitive keypresses. Part-way through each game, we introduced latency and measured how quickly players adapted. We manipulated auditory feedback to be Immediate (upon keypress) or Delayed (synchronized with the visual event). Across both experiments, Immediate auditory feedback substantially accelerated adaptation. We conclude that feedback about when input was registered aids players' implicit "debugging" of in-game latency. Compared to infrastructure solutions to latency such as laying new fiber-optic cable, providing immediate auditory feedback upon keypress presents a low-cost way to improve visuomotor adaptation in online games.

CCS Concepts

• **Human-centered computing** → **Auditory feedback**; **Empirical studies in HCI**.

Keywords

Audio/Video, Games/Play, Desktop/Laptop Computers, Interaction Design, Quantitative Methods

ACM Reference Format:

Xuyuan Duan, Colleen Macklin, and Benjamin F. van Buren. 2026. Immediate Auditory Feedback Speeds Visuomotor Delay Adaptation in Video Games. In *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26)*, April 13–17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3772363.3798405>

1 Introduction

Online and cloud gaming present players with two related timing challenges. First, online systems have inherent baseline latency (commonly 60ms+) between player input and onscreen events, which is substantially higher than the 20-30ms typical of local play [8]. Second, internet network instability causes latency between keypress registration and resulting onscreen events to fluctuate, requiring ongoing recalibration by players to adjust how early they enter their inputs. Overcoming both challenges demands the same underlying ability: players must accurately perceive how much latency is occurring in the system between their input and a resulting onscreen event, and adjust their input timing to be that much earlier. Here we propose that providing immediate auditory feedback when a player's input is registered helps players to perceive more precisely when their input was detected by the system, which in turn allows more precise measurement of latency arising in the game and speeds adaptation to it. In other words, we hypothesize that giving players immediate auditory input feedback when their input is registered aids their implicit "debugging" of how much latency is arising following keypress registration, in the game itself, and thus speeds adaptation to visuomotor latency when it arises.



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ACM ISBN 979-8-4007-2281-3/26/04
<https://doi.org/10.1145/3772363.3798405>

To test this hypothesis, we designed two browser-based games requiring time-sensitive responses, and for each, ran a preregistered experiment in which participants played one of two versions of the game. Partway through, we introduced latency between players' inputs and onscreen events, and we measured how quickly participants adapted to this by entering inputs earlier. Critically, in one version of the game participants received Immediate auditory feedback at the moment their input was registered, and in the other they received Delayed auditory feedback which matched the timing of the delayed onscreen event. To foreshadow, in both experiments, immediate auditory feedback accelerated visuomotor latency adaptation. We propose this as a low-cost design intervention for online games and other latency-prone systems.

2 Related Work

Latency degrades performance in interactive systems. For example, in a target acquisition task using a mouse, 225ms of latency increased movement time by 64% and error rates by 214% [11]. More recent work has established finer thresholds: users can detect latency as low as 25ms when tapping a touchscreen and observing resulting onscreen events [6]. Even sub-125ms reductions yield measurable performance benefits in first-person shooters [9]. When playing video games on the Cloud, performance degradation due to latency is usually unavoidable. When players interact with cloud gaming services, the delay from keypress to onscreen response ranges from 65-130ms, roughly triple that of local play (22-44ms) [8]. The gap widens for users physically more distant from data centers [3]. The games industry has developed various techniques to mask latency from players [2, 10]. However, these approaches are inappropriate for timing-sensitive tasks, in which players need to precisely measure and adapt to latency by pressing keys earlier. There has been little work considering (let alone empirically testing) design interventions to support visuomotor adaptation to latency in games.

When latency occurs in a game (e.g. due to fluctuating internet connectivity), to visuomotorically adapt to this, the player must determine how much of the total latency arose after their input was registered [1]. Without feedback at the moment of input registration, players have only an imprecise representation of when their keypress was received — due to the relatively poor temporal precision of haptic perception [7, 14], and to the fact that when pressing a key, one cannot know precisely when during its mechanical depression the keypress was actually registered. This imprecision propagates to their representation of the latency itself, since the amount of latency is the interval between when the keypress was registered and when the onscreen feedback occurs.

3 The Present Study

We hypothesize that receiving feedback on input registration timing is critical for players to adapt to changes in latency in a game system. Given that the latency players must adapt to is the interval between when their input is registered and the resulting onscreen event, we reasoned that providing *immediate auditory feedback* upon keypress might provide a precise temporal marker of when input is registered, enabling a more precise representation of the subsequent latency which requires adapting to. (We focus on

auditory feedback for two reasons: First, auditory stimuli are perceived with greater temporal precision than visual or haptic stimuli [7, 14, 17]. Second, auditory feedback leverages sound effects already common in games, minimizing disruption to immersion and avoiding competition for visual attention during gameplay.) However, if feedback is not provided, or if auditory feedback is delayed along with the delayed visual event, then players must rely on a more imprecise representation of keypress registration timing to measure and adapt to the delay which has been introduced in the system.

To test this hypothesis, we designed two browser-based games in which performance depended on precise input timing. In both games, players played for a short time with no visuomotor latency between their keypresses and onscreen events, after which a small latency was introduced. For each game, we ran a between-participants experiment in which we varied the timing of auditory feedback to be Immediate upon keypress (for half of participants) or Delayed with the delayed visual event (for the other half). We predicted that, following the introduction of latency, Immediate (vs. Delayed) auditory feedback would cause players to adapt faster by entering inputs earlier.

4 Task Design

4.1 Game 1: Interception

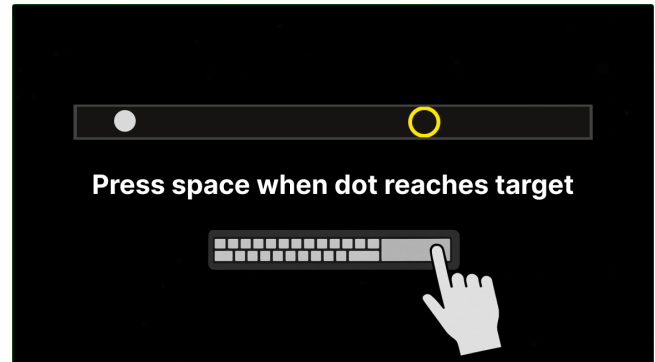


Figure 2: The Interception task. A dot moves rightward toward a randomly horizontally positioned target circle. Players press the spacebar to stop the dot, earning points based on proximity to the target's center. Optimal performance requires anticipating latency between keypress and visual response.

We first tested the effect of Immediate vs. Delayed auditory feedback on delay adaptation in a simple, custom built Interception game, in which players pressed a key to stop a moving dot as close as possible to a target (see Figure 2). We designed this task to be simple enough that adaptation did not depend on learning complex game mechanics, while still requiring the precise timing characteristic of popular action games. On each trial, a 20px diameter dot began at the left side of a 720px × 40px rectangle and moved rightward toward a 28px target circle positioned randomly between 50% and 80% of the distance from the left edge of the rectangle. Each trial lasted for 1.3 seconds, with 0.5s wait time before the dot started to

move and 0.8s for the dot to reach the right edge of the rectangle. Participants pressed the spacebar to stop the dot as close as possible to the circle's center and received more points the closer they were to the center. In this game, stopping near the circle's center requires pressing the key before the dot reaches the target, with the optimal lead time depending on any latency between one's keypress and the onscreen stopping event.

4.2 Game 2: Modified DinoRun

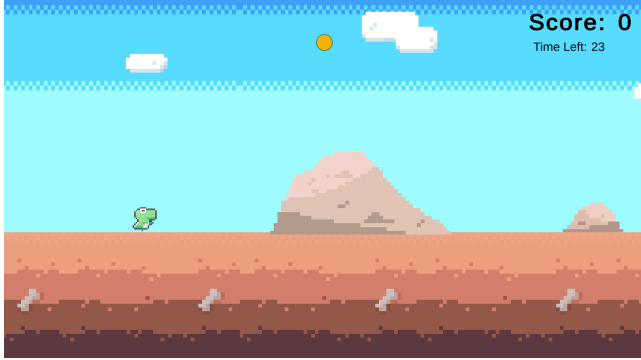


Figure 3: Modified DinoRun game. Coins appear at jittered intervals and scroll leftward. Players must time their jumps so their avatar reaches peak height when the coin passes, requiring anticipation of any latency between keypress and jump initiation. Visual assets adapted from “Bolt 2D DinoRun Assets Pack” (Goldmetal) under the Unity Asset Store Standard EULA. (<https://assetstore.unity.com/packages/2d/characters/bolt-2d-dinorun-assets-pack-188721>)

To test whether Immediate vs. Delayed auditory feedback speeds delay adaptation in a more complex (and popular) game, we built a variant of DinoRun, the classic Chrome browser game (see Figure 3). Players played in-browser in Unity WebGL. The player's avatar remained fixed at the left side of the screen while the background scrolled leftward, creating the impression that the player's avatar was moving rightward. Hovering coins appeared on the right edge of the display every 3 seconds ($\pm$ a random jitter of up to 450ms) and moved toward the player at the same speed as the background elements. Coins were positioned high up, so players had to press their key to initiate jumps early, so their avatar reached the peak of its jump at the moment the coin passed by in order to collect the coin. We retained binary scoring (coin collected or missed) to preserve ecological validity with real platformer games, where outcomes are typically all-or-nothing.

5 Experiments

We conducted two preregistered experiments using these games (Experiment 1: Interception, $N=400$; Experiment 2: DinoRun, $N=300$). Both experiments followed the same general procedure, varying only the game, the latency magnitude, and the number of post-latency trials.

5.1 Procedure

Participants were recruited from Prolific (<https://www.prolific.com/>) and completed the experiment in their web browser after passing an audio check. Each experiment consisted of three phases: Practice (10 trials with no latency), Baseline (10 trials with no latency), and Latency (20 trials for Interception, 30 trials for DinoRun). During the Latency phase, we secretly introduced latency between keypress and visual response (120ms for Interception, 200 ± 33 ms for DinoRun).

In the Practice and Baseline phases, an immediate beep sound (100ms, 200 Hz) accompanied input during the practice and baseline phases. For the half of participants in the Immediate Auditory Feedback condition, during the Latency period the sound continued to play instantly upon keypress even though there was latency between the keypress and the onscreen event. For participants in the Delayed Auditory Feedback condition, during the Latency period the sound was delayed to be synchronous with the delayed visual event.

5.2 Measures

As per our preregistration, on each trial, we measured the participant's lead time — i.e. how early they entered input relative to the moment that their dot (dinosaur) reached the circle (coin). We defined a player's Adaptation Latency as the first post-latency trial index i when the absolute difference between their current lead time L_i and their 'target' lead time — their mean lead time across trials in the Latency phase $\mu_{\text{latency_trials}}$ was $\leq \frac{1}{3}$ of the initial error E_{initial} , where $E_{\text{initial}} = |L_1 - \mu_{\text{latency_trials}}|$.

$$\text{Adaptation Latency} = \min\{i : |L_i - \mu_{\text{latency_trials}}| \leq \frac{1}{3} E_{\text{initial}}\}$$

5.3 Results

5.3.1 Experiment 1 (Interception). Inspection of Figure 4A suggests that the Immediate Auditory Feedback group adapted faster than the Delayed Auditory Feedback group. This is indicated in the top (line) graph by a steeper curve following delay onset, and in the bottom (bar) graph by an earlier Adaptation Index in the Immediate ($M=4.99$ trials post-latency, $SD=4.74$) vs. Delayed ($M=6.48$ trials post-latency, $SD=6.61$) condition, $t(398)=2.59$, $p=0.010$, $d=0.26$. In addition, Immediate Feedback participants showed smaller post-latency score drops ($M=9.13\%$) compared to Delayed Feedback participants ($M=10.58\%$), $t(398)=2.22$, $p=0.027$, $d=0.22$.

5.3.2 Experiment 2 (DinoRun). Inspection of Figure 4B suggests that the Immediate Auditory Feedback group again adapted faster than the Delayed Auditory Feedback group. This is indicated in the top graph by the steeper curve following delay onset, and in the bottom graph by the earlier Adaptation Index in the Immediate ($M=4.86$ trials post-latency, $SD=3.93$) vs. Delayed ($M=6.30$ trials post-latency, $SD=5.26$) condition, $t(298)=2.69$, $p=0.008$, $d=0.31$. Unlike Experiment 1, post-latency score drops did not differ between conditions ($p=0.838$).

6 Discussion

Across two experiments, immediate (vs. delayed) auditory feedback consistently accelerated visuomotor adaptation to latency between

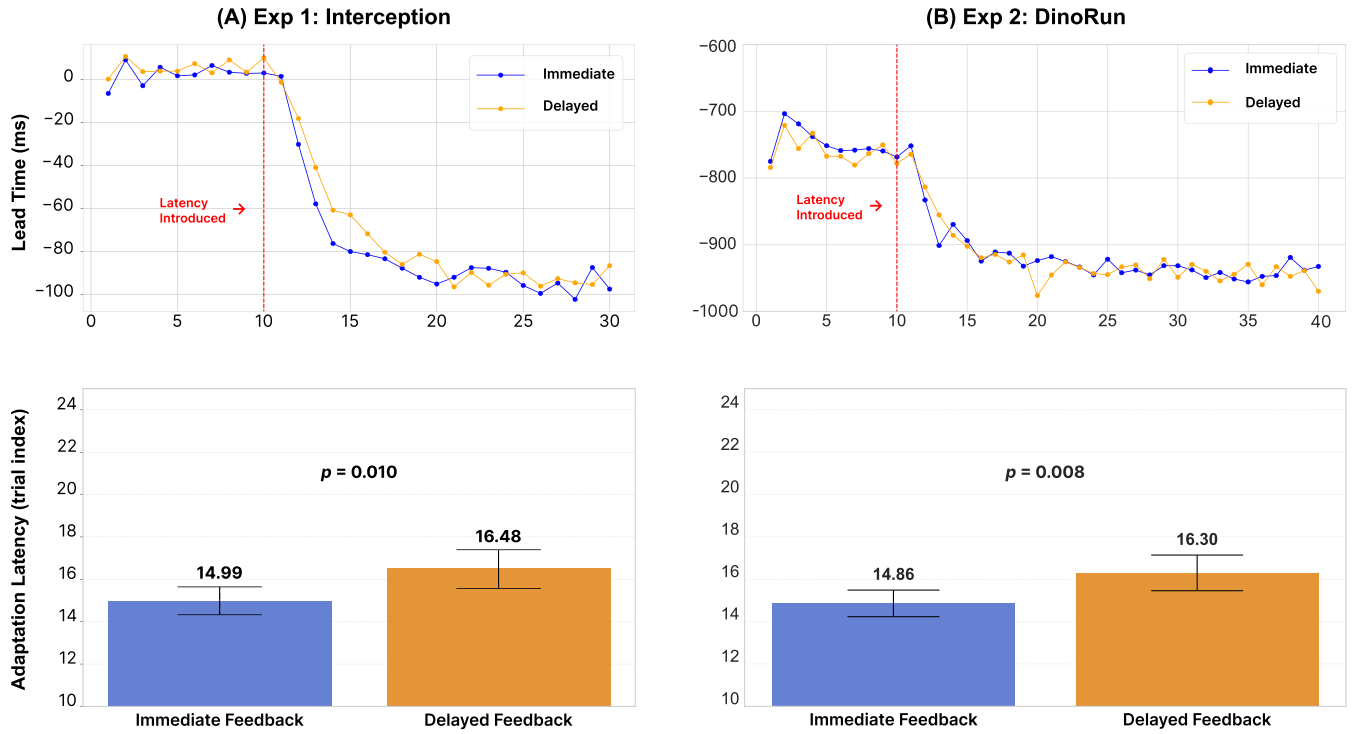


Figure 4: Effects of Immediate vs. Delayed auditory feedback on motor adaptation following the onset of latency between inputs and onscreen events. (A) Results of Interception experiment. (B) Results of DinoRun experiment. In both experiments, Immediate Auditory Feedback accelerated adaptation following delay onset compared to Delayed Auditory Feedback.

input registration and onscreen events. These results were highly robust: both experiments were preregistered and well powered, with a combined total of 700 participants. They were also quite general, arising in different tasks (a simple interception game, and a popular platformer) with different latency magnitudes and ranges (a fixed latency of 120ms, and a variable latency $[200 \pm 33\text{ms}]$).

6.1 Theoretical Interpretation

These results suggest that immediate auditory feedback provides a precise temporal marker of input registration, transforming latency into a directly perceivable interval (from the beep to the onscreen event). By contrast, when auditory feedback is delayed, players must measure the interval between a more temporally imprecise representation of when they pressed the key and the onscreen event — resulting in a relatively imprecise measurement of latency, and relatively slow adaptation. In other words, if latency can arise at several steps from initiating a motor command to seeing an onscreen event, providing auditory feedback aids players in implicitly “debugging” what portion of latency is arising in the game, between the registration of their keypress and the onscreen event. This is analogous to a developer isolating which component in a pipeline is causing latency. Immediate auditory feedback provides a “breakpoint” that marks input registration, allowing players to attribute subsequent delay to the game system rather than to uncertainty about when input was registered.

Whereas programmers use breakpoints to slowly and deliberately diagnose where in a system latency is arising, in the present experiments, we suspect that participants’ debugging of when in the chain of events latency was arising was largely implicit. First, past work has often found evidence of visuomotor adaptation without conscious awareness [12, 16]. Second, in debriefing responses, participants rarely mentioned using explicit strategies to update their input timing. However, there were a couple of interesting exceptions: e.g. one participant who received Immediate auditory feedback reported that this aided their motor adaptation, writing, “The audible sound...helped me discern the time between the button press and the ball stopping.” In contrast, one participant who received Delayed feedback complained, “I attempted to stop my keypresses earlier; however, I struggled to decide how much earlier to press.” These comments suggest that immediate feedback not only speeds motor adaptation, but may (for some players) also improve player experience, consistent with prior work on game feel [4, 15].

6.2 Design Implications

Game design heuristics often emphasize the importance of providing players with immediate, consistent feedback in response to their inputs [4, 13]. The present study demonstrates a new and specific benefit of immediate feedback: it speeds adaptation to latency when it occurs. To ensure that players quickly adapt to latency between input registration and onscreen events, we recommend

always providing immediate auditory confirmation of input registration. In practice, this need not require adding new sounds: many games already play action-triggered audio (e.g., a jump sound in a platformer, a gunshot in a first-person shooter) that could serve as the temporal anchor, so long as these sounds are triggered at input registration rather than synchronized with the slightly delayed visual outcome. Here we have focused on PC video games; but we expect that this technique may be applied broadly, e.g. to facilitate adaptation to rendering latency in VR/AR systems, as well as latency in remote collaboration tools, digital musical instruments [5], and telemedicine interfaces. The key insight is that when the onscreen effects of our actions cannot be immediate, providing auditory confirmation of input registration helps users to calibrate their input timing. Immediate feedback may prove particularly valuable under fluctuating network conditions, where players must continuously recalibrate rather than adapt just once — but this remains to be tested. Even small improvements in adaptation speed may compound over extended play sessions where players repeatedly encounter latency fluctuations.

Immediate auditory feedback sped visuomotor adaptation in both experiments, and even reduced players' post-latency score drop in the Interception game, in which points were awarded based on proximity to the target circle. That immediate auditory feedback did not reduce the post-latency score drop in DinoRun may simply reflect its binary scoring system (coin collected/missed), wherein a small improvement in visuomotor adaptation speed might sometimes bring one closer to touching coins without actually collecting them and earning points. Thus, under conditions of latency, immediate auditory feedback may produce more reliable score improvements when score is a continuous function of how close the player comes to a target. Moreover, quite independently of auditory feedback timing, continuous scoring systems provide players with feedback about how close they got to their target — which may aid visuomotor adaptation when latency occurs.

7 Conclusion

Across two preregistered experiments using different games and latency amounts, immediate auditory feedback substantially accelerated visuomotor adaptation to latency. Feedback about when input was registered helps players implicitly “debug” what portion of the latency between their keypress and an onscreen event is happening in the game; helping players to more precisely measure how much latency is arising in the game in turn speeds up how quickly they adapt to it. Latency in online and cloud games is a problem which cannot be fully solved. Compared to infrastructure improvements like laying fiber-optic cable, providing immediate auditory feedback at input registration presents a low-cost design intervention that can improve player performance in online games and other interactive systems subject to latency.

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A Research Methods

All experimental procedures and analyses were preregistered prior to data collection (Experiment 1: <https://osf.io/e2wm4>; Experiment 2: <https://osf.io/s3pjb>). We chose participants' mean lead time across latency-phase trials as the adaptation target because it provides a model-free estimate that does not assume a specific functional form (e.g., exponential decay) for the adaptation curve. While alternative approaches such as fitting time constants or using final-trial asymptotes are viable, they introduce additional modeling assumptions; the present metric was selected for its simplicity and directness.